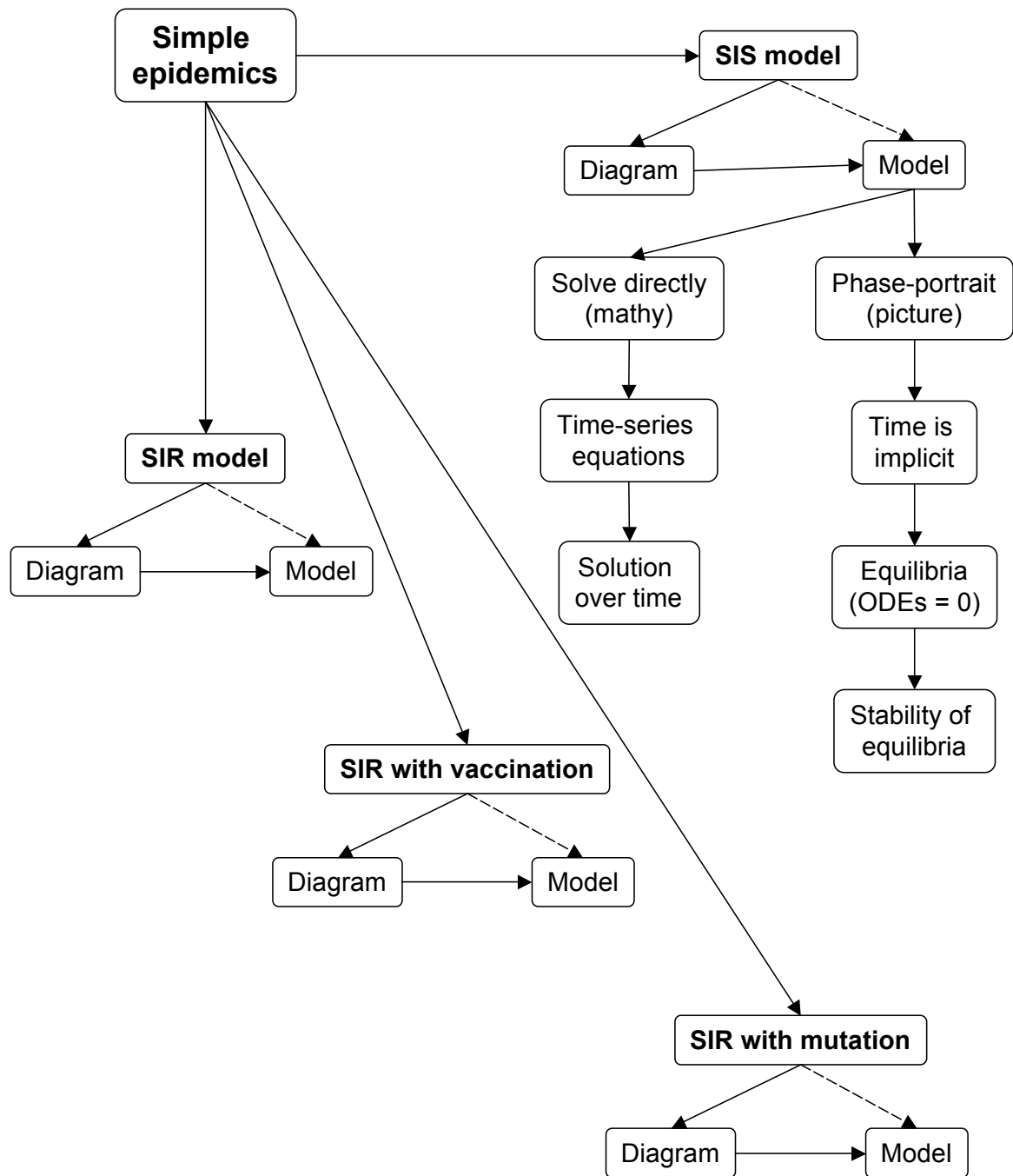


Simple epidemic models

- Construct ODE (Ordinary Differential Equation) models
- Relationship between the diagram and the equations
- Alter models to include other factors.



Ordinary Differential Equations(ODEs)

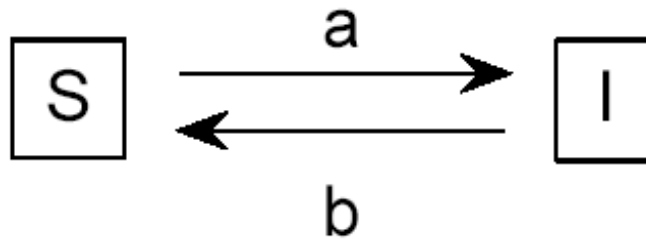
- ODEs deal with populations, not individuals
- We assume the population is well-mixed
- We keep track of the inflow and the outflow.

SIS epidemic

- Susceptible $\rightarrow$ Infected $\rightarrow$ Susceptible
- You get sick, then recover, but without immunity
- E.g. the common cold.

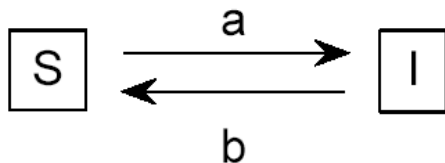
Diagram

- Susceptibles become infected at rate a
- Infecteds recover at rate b .



SIS equations

- Becoming infected depends on contact between Susceptibles and Infecteds (aSI)
- Recovery is at a constant rate, proportional to number of Infecteds (bI).



$$\begin{aligned}\frac{dS}{dt} &= bI - aSI \\ \frac{dI}{dt} &= aSI - bI\end{aligned}$$

a = infection rate
 b = recovery rate

Total population is constant

- Add equations together
- $N=S+I$ (total population)
- $dN/dt=0 \rightarrow N$ is a constant.

$$\frac{dS}{dt} + \frac{dI}{dt} = bI - aSI + aSI - bI$$

$$\frac{dN}{dt} = 0$$

$$\begin{aligned} S' &= bI - aSI \\ I' &= aSI - bI \end{aligned}$$

S = Susceptible
I = Infected

Solving directly

- Since $N=S+I$, this means $S = N-I$

$$\begin{aligned}\frac{dI}{dt} &= a(N - I)I - bI \\ &= (aN - b - aI)I\end{aligned}$$

$$I' = aSI - bI$$

- Let $A = aN - b$ be a constant

$$\frac{dI}{dt} = (A - aI)I.$$

S = Susceptible
I = Infected
a = infection rate
b = recovery rate

Separate the variables

- Put the I 's on one side and the t 's on the other
(including dI and dt)

$$\frac{dI}{(A - aI)I} = dt .$$

I = Infected
 a = infection rate
 $A = aN - b$ (constant)
 N = total pop.
 b = recovery rate

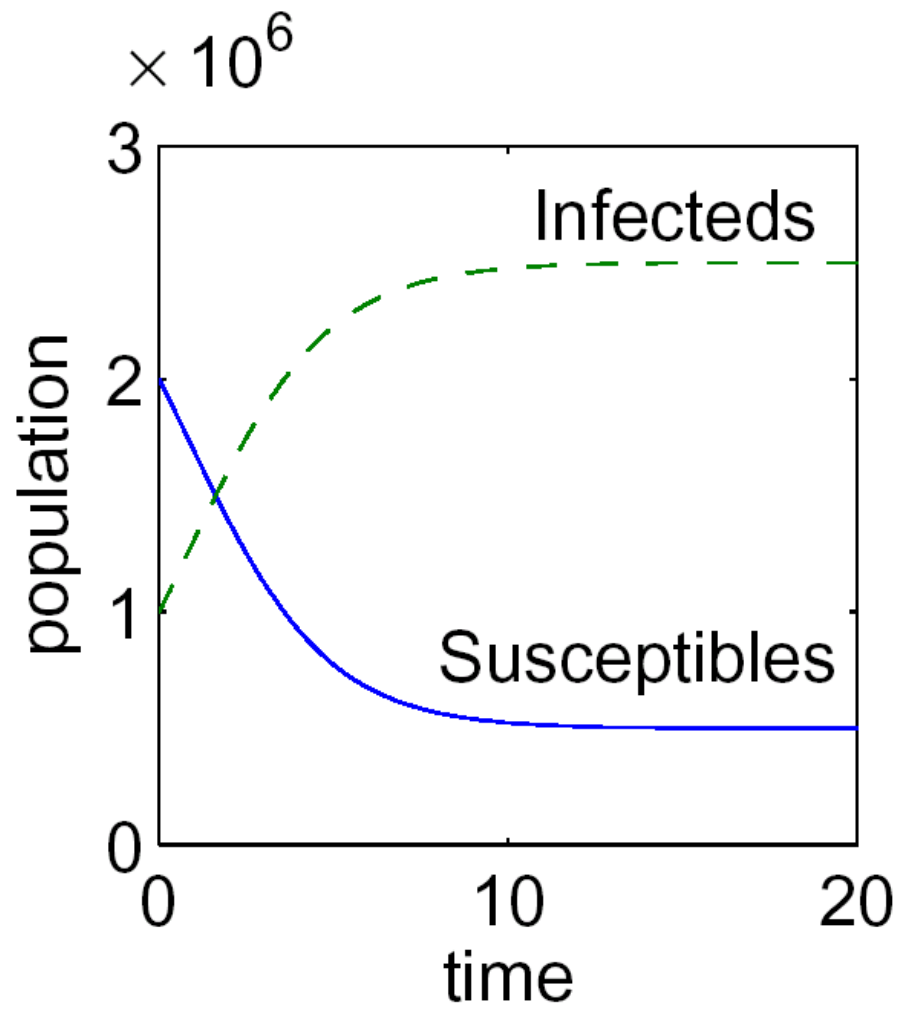
Time series solution

- Rearrange using partial fractions
- Integrate
- Use initial condition $I(0)=I_0$

S = Susceptible
I = Infected
a = infection rate
N = total pop.
b = recovery rate

$$I = \frac{(aN - b)I_0 e^{(aN-b)t}}{(aN - b) + aI_0[e^{(aN-b)t} - 1]}$$

(See
Epidemic
Notes)

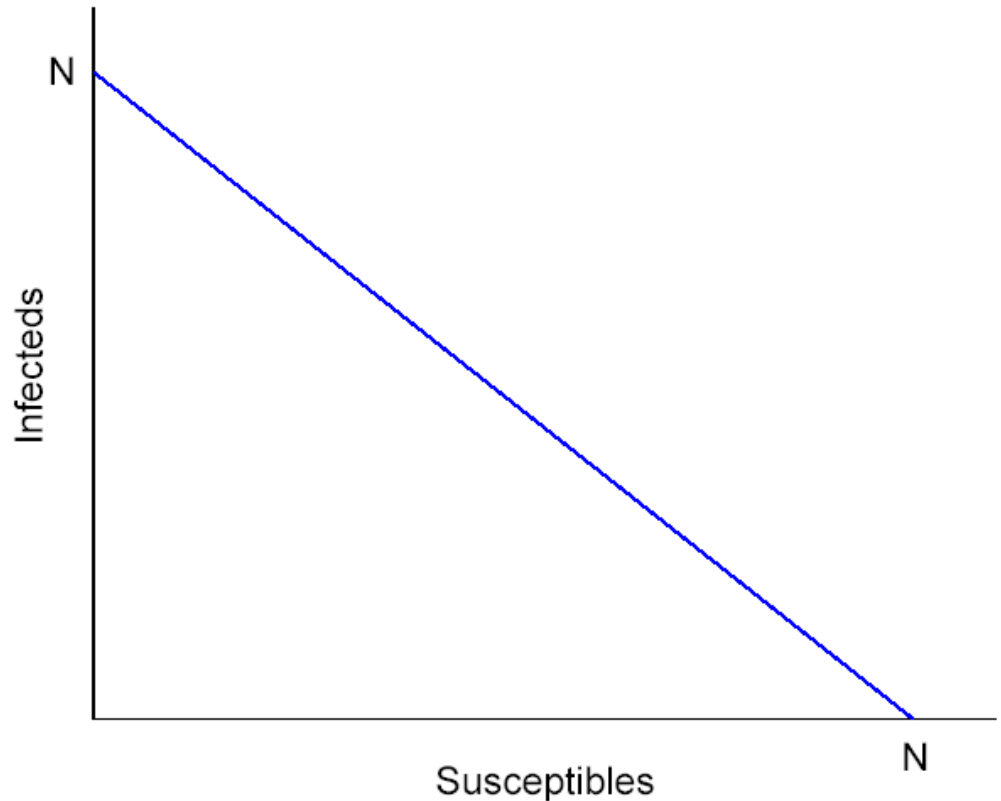


$b=0.1$

$b = \text{recovery rate}$

Phase portraits

- Since $N = S + I$,
 $I = -S + N$
- This is a straight line in I and S
- Time is implicit.



S = Susceptible
I = Infected
N = total pop.

Equilibrium points

- Equilibria occur when derivatives are zero:

$$(b - aS)I = 0 \quad \text{when } S = \frac{b}{a} \text{ or when } I = 0$$

$$(aS - b)I = 0 \quad \text{when } S = \frac{b}{a} \text{ or when } I = 0$$

(We'll call $\frac{b}{a}$ 'p'.)

S = Susceptible
I = Infected
a = infection rate
b = recovery rate

$S' = bI - aSI$
 $I' = aSI - bI$

Two equilibria

- Thus our equilibrium points are

$$(\bar{S}, \bar{I}) = (p, N - p)$$

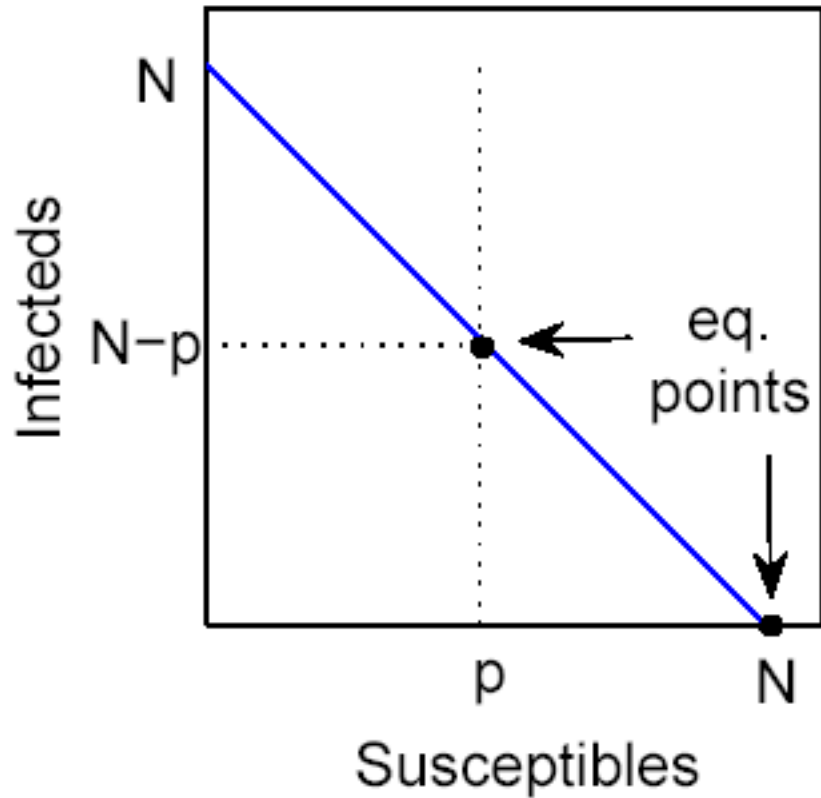
or

$$(\bar{S}, \bar{I}) = (N, 0)$$

- The latter always exists, the former is only biologically reasonable if $p < N$.

<i>S = Susceptible</i>	<i>I = Infected</i>
<i>N = total pop.</i>	<i>p = b / a</i>
<i>b = recovery rate</i>	<i>a = infection rate</i>

Case I: $p < N$



N = total pop. $p = b / a$
 b = recovery rate a = infection rate

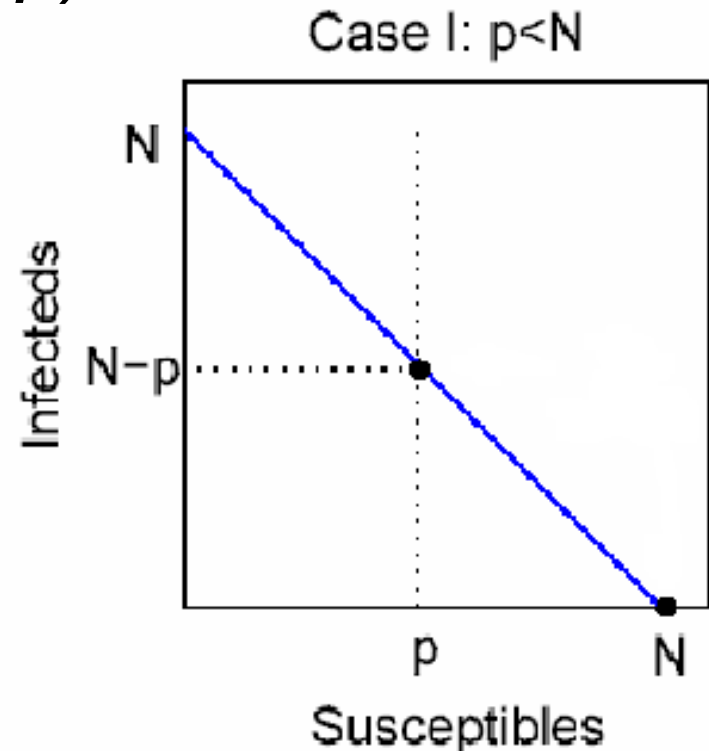
Stability

$$S' = a(p-S)I$$

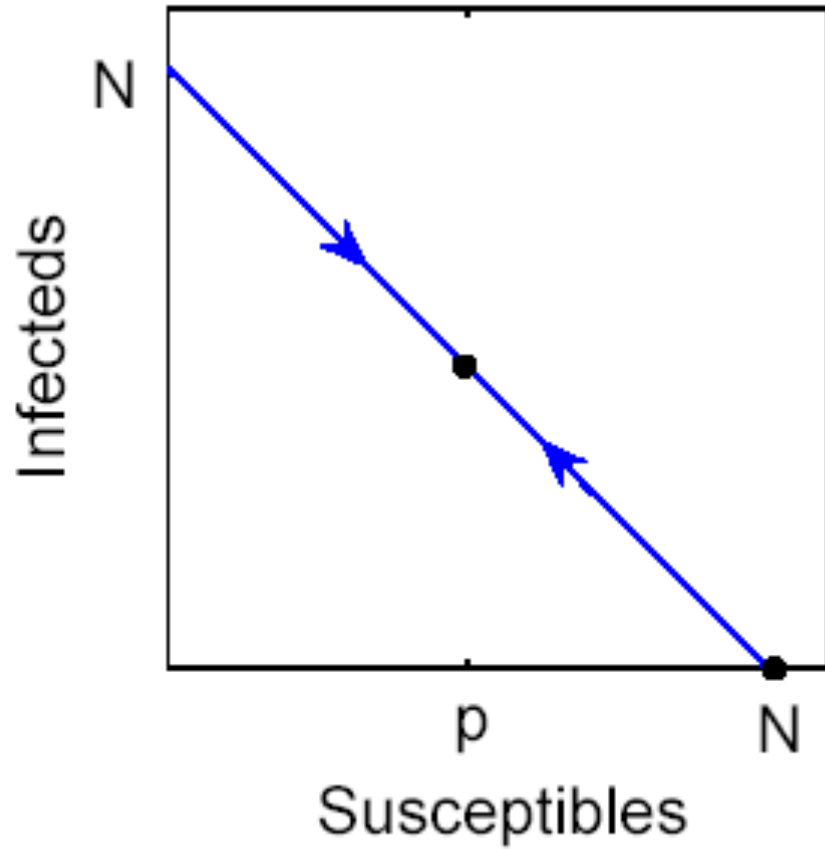
$$I' = a(S-p)I$$

- $S < p \rightarrow S' > 0, I' < 0$
- $S > p \rightarrow S' < 0, I' > 0$

S = Susceptible	I = Infected
N = total pop.	$p = b / a$
b = recovery rate	a = infection rate



Case I: $p < N$

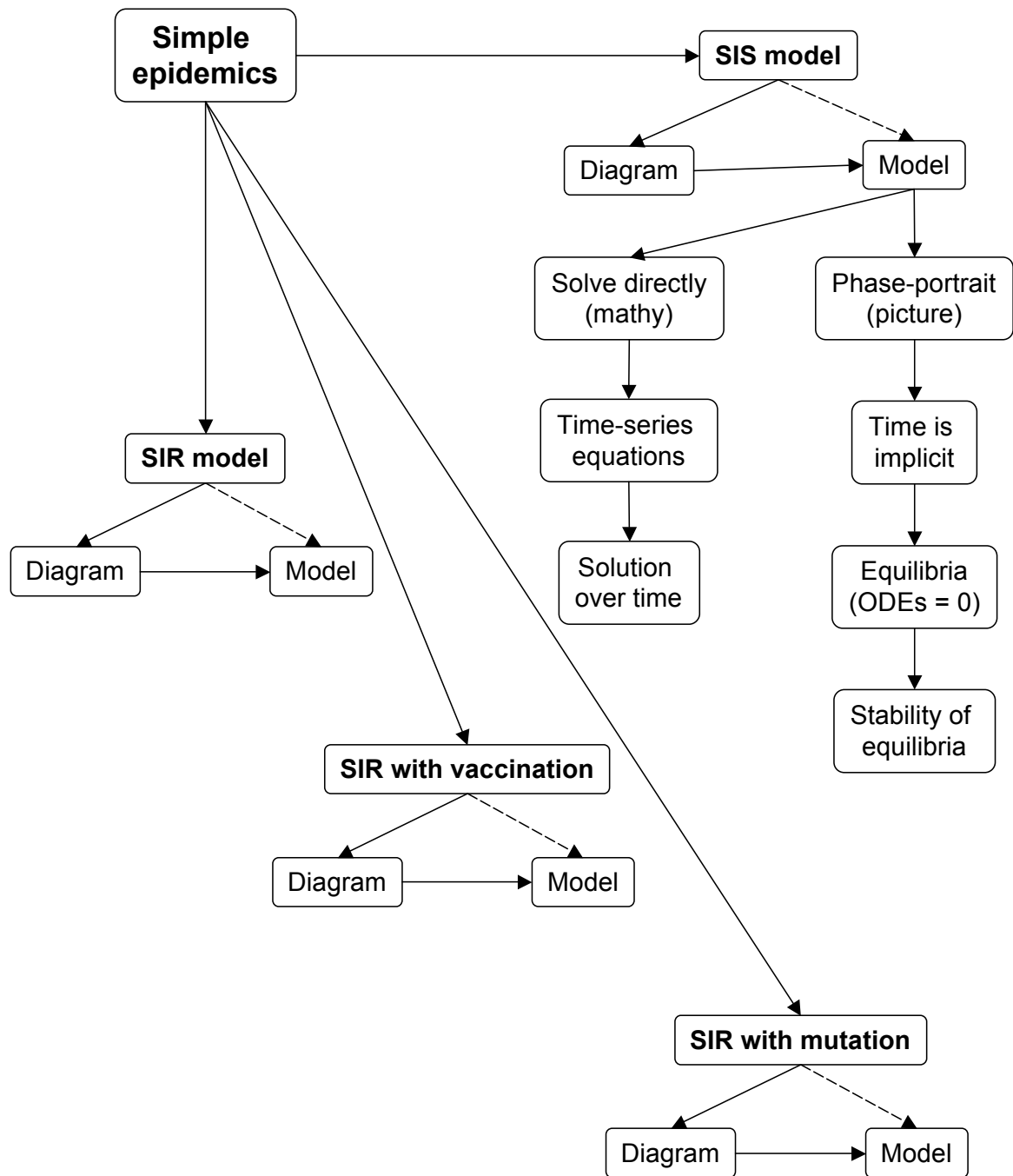


N = total pop. $p = b / a$
 b = recovery rate a = infection rate

Stability implications

- When $p=b/a > N$, the recovery rate is high, so infecteds recover quickly and the population moves to a population of susceptibles
- When $p=b/a < N$, the infection rate is high and the infection stabilises at an endemic equilibrium.

N = total pop.
b = recovery rate
a = infection rate



SIR epidemics

- Susceptible $\rightarrow$ Infected $\rightarrow$ Removed
- Removed can be recovered, immune, or dead.



a = infection rate
b = recovery rate

SIR equations

- Becoming infected depends on contact between Susceptibles and Infecteds (aSI)
- Recovery is at a constant rate, proportional to number of Infecteds (b).

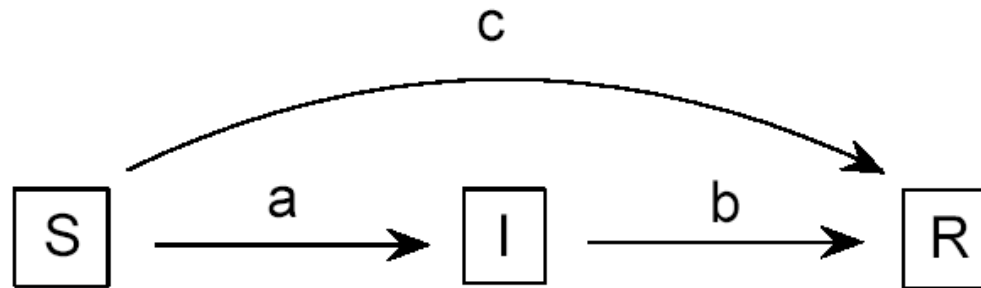


a = infection rate
 b = recovery rate

$$\begin{aligned}\frac{dS}{dt} &= -aSI \\ \frac{dI}{dt} &= aSI - bI \\ \frac{dR}{dt} &= bI\end{aligned}$$

SIR with vaccination

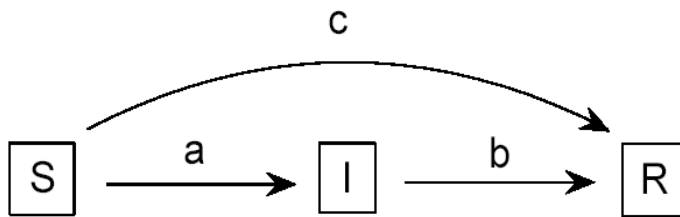
- A vaccine sends some Susceptibles directly to the Recovered (immune) state
- $N=S+I+R$.



<i>S = Susceptible</i>	<i>I = Infected</i>
<i>R = Recovered</i>	<i>a = infection rate</i>
<i>b = recovery rate</i>	<i>c = vaccination rate</i>

Vaccination equations

- Vaccination is assumed to be a fixed number of shots per time period (c).

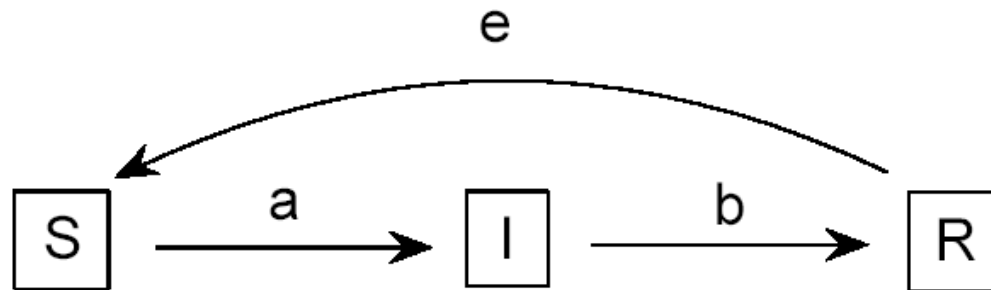


S = Susceptible
I = Infected
R = Recovered
a = infection rate
b = recovery rate
c = vaccination rate

$$\begin{aligned}\frac{dS}{dt} &= -aSI - c \\ \frac{dI}{dt} &= aSI - bI \\ \frac{dR}{dt} &= bI + c\end{aligned}$$

SIR with mutation

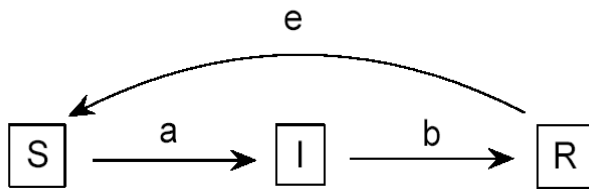
- If the virus mutates, Recovereds lose their immunity.



<i>S = Susceptible</i>	<i>I = Infected</i>
<i>R = Recovered</i>	<i>a = infection rate</i>
<i>b = recovery rate</i>	<i>e = mutation rate</i>

Mutation equations

- A time-delay T allows a 'grace period' before people are susceptible again
- They become susceptible at a rate (e) depending on their status at time $t-T$.



S = Susceptible
I = Infected
R = Recovered
a = infection rate
b = recovery rate

$$\frac{dS(t)}{dt} = -aS(t)I(t) + eR(t - T)$$

$$\frac{dI(t)}{dt} = aS(t)I(t) - bI(t)$$

$$\frac{dR(t)}{dt} = bI(t) - eR(t - T)$$

Delay Differential Equations

- These are called delay-differential equations
- They are harder to analyse than ordinary differential equations, but are often more realistic.